

SDA Vision / Research report

earth-image.jpg



Assessment summary

Field	Recorded detail
Finding	Leaning authentic
Recorded verdict	authentic likely
Confidence	medium
Indicative rating	5/100
Rating basis	median of available model ratings; not a calibrated probability
Explanation	3 models lean authentic with matching verdicts and ratings. Review the other findings, original source and context alongside this result.
Coverage	image - 1 frame(s) analysed
Review context	Research assessment. Review alongside source information and context. When credentials are missing, origin remains unresolved by this check. Use these findings to inform a documented human review.

Provider overview

Provider	Type / status	Assessment	Rating
Claude Vision	vision / ok	authentic likely	5/100
OpenAI Vision	vision / ok	authentic likely	2/100
Gemini Vision	vision / ok	authentic likely	15/100
C2PA Content Credentials	provenance / ok	inconclusive	n/a
Local forensic cues	forensic / ok	partially synthetic	47/100

Provider evidence / 1

Field	Recorded detail
Provider	Claude Vision
Model	claude-opus-5-5
Status	ok
Assessment	authentic likely
Confidence	high
Role in assessment	Supporting
Provider summary	synthetic likelihood 5/100; tells: composition matches the widely circulated 1972 Apollo 17 'Blue Marble' photograph (AS17-148-22727), geographically accurate Africa, Arabian Peninsula, Red Sea, Gulf of Aden and Madagascar outlines, Antarctic ice cap consistent with December solstice illumination, physically coherent cloud structures: cyclonic swirls, frontal bands, open-cell cumulus fields
Evidence 1	Composition matches the widely circulated 1972 Apollo 17 'Blue Marble' photograph (AS17-148-22727)
Evidence 2	Geographically accurate Africa, Arabian Peninsula, Red Sea, Gulf of Aden and Madagascar outlines
Evidence 3	Antarctic ice cap consistent with December solstice illumination
Evidence 4	Physically coherent cloud structures: cyclonic swirls, frontal bands, open-cell cumulus fields
Evidence 5	Film-scan grain and slight color cast consistent with analog Hasselblad film digitization
Evidence 6	Natural limb darkening and atmospheric haze at globe edge
Evidence 7	No repeating textures, smeared regions or implausible landmass shapes typical of generative models
Evidence 8	Caveat: Visual match to a famous archival photo is strong, but a high-quality AI recreation or a color-adjusted/re-edited version could look similar; confirm via reverse image search against the NASA archive original and check file metadata.

Provider evidence / 2

Field	Recorded detail
Provider	OpenAI Vision
Model	gpt-6-astra
Status	ok
Assessment	authentic likely
Confidence	high
Role in assessment	Supporting
Provider summary	synthetic likelihood 2/100; tells: Coherent coastlines of Africa, Madagascar and the Arabian Peninsula, Irregular, finely detailed cloud systems with consistent spherical foreshortening, Natural atmospheric haze and gradual shading around Earth's limb, Fine photographic grain and slight color fringing along the globe edge
Evidence 1	Coherent coastlines of Africa, Madagascar and the Arabian Peninsula
Evidence 2	Irregular, finely detailed cloud systems with consistent spherical foreshortening
Evidence 3	Natural atmospheric haze and gradual shading around Earth's limb
Evidence 4	Fine photographic grain and slight color fringing along the globe edge
Evidence 5	Caveat: Resembles the historical Blue Marble photograph; source-file comparison is needed to establish provenance and exclude subtle edits.

Provider evidence / 3

Field	Recorded detail
Provider	Gemini Vision
Model	gemini-3.8-flash
Status	ok
Assessment	authentic likely
Confidence	high
Role in assessment	Supporting
Provider summary	synthetic likelihood 15/100; tells: authentic Apollo 17 'Blue Marble' photograph, natural film grain structure, consistent atmospheric limb glow, accurate meteorological cloud swirls
Evidence 1	Authentic Apollo 17 'Blue Marble' photograph
Evidence 2	Natural film grain structure
Evidence 3	Consistent atmospheric limb glow
Evidence 4	Accurate meteorological cloud swirls
Evidence 5	Caveat: High-profile historic images are frequently duplicated, cropped, or color-adjusted across digital archives.

Content Credentials / provenance

Field	Recorded detail
Provider	C2PA Content Credentials
Model	Not recorded
Status	ok
Assessment	inconclusive
Confidence	Not recorded
Role in assessment	Additional finding
Provider summary	Embedded credentials: none found. Origin remains unresolved by this check.
Evidence 1	Coverage: credentials embedded in the submitted file. External credentials were not retrieved.
Evidence 2	Verifier: c2pa-python 0.37.12; native SDK 0.91.0; specification target 2.4; policy embedded-offline-reviewed-trust-v3.
Evidence 3	Offline embedded-manifest verification; no network trust-list, manifest or revocation refresh.
Evidence 4	No reviewed local C2PA trust snapshot installed. Provenance overrides are disabled.

Provider evidence / 5

Field	Recorded detail
Provider	Local forensic cues
Model	Not recorded
Status	ok
Assessment	partially synthetic
Confidence	low
Role in assessment	Additional finding
Provider summary	Heuristic synthetic likelihood 47/100 from metadata + pixel statistics.
Evidence 1	No camera EXIF (common in AI exports, screenshots, and re-saves).
Evidence 2	ELA mean 0.5, peak 10 (uniform high ELA can indicate generation/heavy edit).
Evidence 3	High-frequency noise std 4.61 (very low/very uniform noise is a synthesis cue).
Evidence 4	Spectral peakiness 25.63 (periodic frequency peaks can mark generated textures).

Evidence supporting this verdict

Field	Recorded detail
Claude Vision	synthetic likelihood 5/100; tells: composition matches the widely circulated 1972 Apollo 17 'Blue Marble' photograph (AS17-148-22727), geographically accurate Africa, Arabian Peninsula, Red Sea, Gulf of Aden and Madagascar outlines, Antarctic ice cap consistent with December solstice illumination, physically coherent cloud structures: cyclonic swirls, frontal bands, open-cell cumulus fields
OpenAI Vision	synthetic likelihood 2/100; tells: Coherent coastlines of Africa, Madagascar and the Arabian Peninsula, Irregular, finely detailed cloud systems with consistent spherical foreshortening, Natural atmospheric haze and gradual shading around Earth's limb, Fine photographic grain and slight color fringing along the globe edge
Gemini Vision	synthetic likelihood 15/100; tells: authentic Apollo 17 'Blue Marble' photograph, natural film grain structure, consistent atmospheric limb glow, accurate meteorological cloud swirls

Additional findings

Field	Recorded detail
Local forensic cues	Supporting observation; considered alongside the model assessments.
C2PA Content Credentials	Embedded credentials: none found. Origin remains unresolved by this check.

How this assessment was reached

Field	Recorded detail
1. override	outcome=none; detail=No confirmed C2PA AI assertion or SynthID watermark. Origin remains unresolved by these checks.
2. model_agreement	synthetic=[]; authentic=['Claude Vision', 'OpenAI Vision', 'Gemini Vision']; uncertain=[]; forensic_supports_synthetic=False
3. verdict	value=authentic_likely; confidence=medium; reason=3 models lean authentic with matching verdicts and ratings. Review the other findings, original source and context alongside this result.

File and analysis record

Field	Recorded detail
File	earth-image.jpg
Analysis ID	b8bb2ceb-20cc-4afd-a616-7cbaace7f08a
Generated	2026-09-26T17:54:47+00:00
Tool	SDA Vision 0.1.0
Models	claude-opus-5-5, gemini-3.8-flash, gpt-6-astra
Method version	verdict-consistency-2026-09-26.1
Analysis lens	general
Original SHA-256	a4bad2aba040fd992aff7483eee6166943e232061b82d7149b2060786719753b
Ingestion note 1	Recomputed locally from the same saved provider replies acquired under run five-examples-20260926-r1. No new provider requests. Original report 6201ac7a-3540-405f-b234-0346b77f7672 was generated at 2026-09-26T17:38:41+00:00 under method publication-repair-2026-09-24.3; that report is preserved.
Ingestion note 2	Method verdict-consistency-2026-09-26.1: directional model votes require matching explicit verdicts and rating thresholds. Inconclusive replies retain their ratings and remain undecided observations.
Ingestion note 3	Presentation copy of saved report cf784e2f-0554-4bed-8880-c825ae7b8282. The descriptive filename replaces the earlier example ID for navigation. Original analysis dates, provider replies, scores and acquisition context are retained; no new model analysis.